

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1774

North Tifton 100 kV
May 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1774. The Interconnection Customer proposes a 100.8 MW solar photovoltaic facility interconnecting at the North Tifton 100 kV substation in the Indiana area. The request was received on June 08, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 631 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,329,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Don – Monroeville 100 KV	Dutton Pump – Leyendecker 100 KV	128.4	113	5.4
Hato Rey Gis 1403-1404 – Don 100 KV	Leyendecker – Dutton Pump 100 KV	111.3	145	5.4
Edes Tap 1 – Hato Rey Gis 1403-1404 100 KV	Dutton Pump – Leyendecker 100 KV line	108.8	162	5.4
Ede Tap 1 – Spearma City Substation 100 KV	Dutton Pump Substation – Leyendecker 100 KV	108.3	167	5.4
China – Ponnequin 100 KV line	Montpelier Electric Generating Sta 230/1 kV TX	104.9	250	13.7
Montpelier Electric Generating Sta 230/1kV TX	Montpelier Electric Generating Sta – Wabash 500 KV	101.6	531	19.3
Montpelier Electric Generating Sta 100 kV/1 kV Transformer Bank	Montpelier Electric Generating Sta – Wabash 500 KV	101.6	531	19.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Don – Monroeville 100 KV	Rebuild line with higher-capacity conductor	\$7,784,000	15% (\$1,165,000)
Hato Rey Gis 1403-1404 – Don 100 KV	Rebuild line with higher-capacity conductor	\$1,695,000	15% (\$254,000)
Edes Tap 1 – Hato Rey Gis 1403-1404 100 KV	Reconductor line	\$3,580,000	15% (\$536,000)
Ede Tap 1 – Spearma City Substation 100 KV	Reconductor line	\$4,385,000	15% (\$656,000)
China – Ponnequin 100 KV line	Reconductor line	\$846,000	4% (\$33,000)
Montpelier Electric Generating Station Substation 230/1 kV Auto	Replace transformer with higher-rated unit	\$12,605,000	4% (\$534,000)
Montpelier Electric Generating Station 100-1 kV XFMR	Replace transformer with higher-rated unit	\$3,564,000	4% (\$151,000)
	Total allocated to project		\$3,329,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.